METHOD AND SYSTEM FOR HIGH-BANDWIDTH, FREE-SPACE OPTICAL DATA EXCHANGE BETWEEN MOBILE DEVICES

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Title: METHOD AND SYSTEM FOR HIGH-BANDWIDTH, FREE-SPACE OPTICAL DATA EXCHANGE BETWEEN MOBILE DEVICES

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Abstract

A method and system for establishing a high-bandwidth, short-range, free-space optical data link between two uncoupled mobile electronic devices, such as smartphones or tablets. The system utilizes the screen of a transmitting device as a frame-based optical emitter and the camera of a receiving device as an optical sensor. Data is encoded into a sequence of frames displayed on the transmitter's screen, where each frame represents a large block of parallel data. To ensure reliability in an unguided, variable environment, the system uses large "channel elements"—grouping blocks of physical pixels into logical super-pixels—to compensate for movement, distance, and angular misalignment. A negotiation and synchronization protocol is established using a preliminary handshake (e.g., displaying a QR code) to allow the receiver to lock on to the transmitter's display and calibrate for the data transfer, enabling multi-gigabit-per-second transfer rates for tasks like migrating the entire contents of

a device.

Background of the Invention

Existing methods for short-range data transfer between mobile devices, such as Bluetooth and Near Field Communication (NFC), are limited to relatively low bandwidths. Transferring large amounts of data, such as the entire contents of a smartphone during a device upgrade, can be a slow and cumbersome process, often requiring a connection to a Wi-Fi network or a physical cable. While these methods are convenient for small files, they are inadequate for the massive data capacities of modern devices. There is a need for a method that allows for the rapid, ad-hoc transfer of gigabytes or terabytes of data between two mobile devices in close proximity, without relying on external network infrastructure or physical cables.

Summary of the Invention

The present invention provides a system and method for ultra-high-bandwidth, unguided, free-space optical data transfer between two mobile devices. The system leverages the existing hardware of the devices: the display screen of the transmitting device acts as a massively parallel optical emitter, and the camera of the receiving device acts as the optical sensor.

The method begins with an initiation and synchronization phase. The two devices are brought into proximity, and the transmitting device displays a synchronization pattern, such as a QR code. The receiving device's camera captures this pattern, which contains the necessary parameters to establish a synchronized link, including the dimensions of the data transmission frames, the data rate, and error correction settings.

Once synchronized, the transmitting device begins displaying a rapid sequence of data frames. In each frame, the screen is divided into a grid of large logical pixels, or "channel elements," where each element comprises a large block of physical screen pixels all set to the same color and brightness. This use of large channel elements makes the transmission robust and tolerant to relative motion, changes in distance, and misalignment between the devices. Data is encoded in the color and brightness of these channel elements.

The receiving device's camera captures this sequence of frames at a high frame rate (e.g., 24, 30, or 60 fps). A software module on the receiving device decodes the sequence of frames in real-time, reconstructing the original data stream. This method effectively turns the devices into a high-speed optical data link capable of transferring the entire contents of a mobile device's memory in seconds.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Government and its agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental and non-commercial purposes.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram of the system in operation, showing two mobile devices establishing a free-space optical data link.

FIG. 2 is a diagram illustrating the concept of a "channel element," where multiple physical screen pixels are grouped into one logical data channel.

FIG. 3 is a flowchart of the data transfer process, from the initial handshake to the completion of the data transmission.

Detailed Description of the Invention

Referring to **FIG. 1**, the system involves a Transmitting Mobile Device (101) and a Receiving Mobile Device (102). The transmitting device (101) utilizes its display screen (103) as an optical emitter. The receiving device (102) utilizes its camera (104) as an optical sensor. To initiate a data transfer, the devices are positioned such that the camera (104) has a clear line of sight to the screen (103). A free-space optical link (105) is established between them.

A key innovation of the method is the structure of the data frames displayed on the transmitter's screen. As shown in **FIG. 2**, the screen (103) is not treated as millions of tiny, individual data channels. Instead, it is divided into a smaller grid of large logical "channel

elements" (201). Each channel element (201) is comprised of a block of many physical pixels (202), all of which are set to the same color and intensity. This spatial redundancy creates a robust signal that is easily resolvable by the receiving camera, even with motion blur, imperfect focus, or variations in distance and angle. For example, a high-resolution smartphone screen could be treated as a simple 10x10 grid of these large channel elements, creating 100 parallel data channels that are highly resistant to environmental variables.

The process of data transfer is detailed in the flowchart of **FIG. 3**. The process begins with the user initiating the transfer on both devices (301). The transmitting device displays a synchronization pattern, such as a QR code, on its screen (302). The receiving device's camera captures and decodes this pattern to acquire the link parameters (303). With the link parameters established, the receiving device's software can now correctly interpret the subsequent data frames.

The transmitting device then begins to encode the source data into a sequence of frames (304). Each frame is constructed using the large channel element grid. The transmitting device displays these frames in rapid succession (305). Concurrently, the receiving device's camera captures the sequence of frames (306). A real-time decoding module on the receiving device processes the captured video, converting the sequence of colored grids back into the original data stream (307). The process continues until all data is transferred, and a completion signal is sent (308).

Claims

What is claimed is:

1. A method for high-bandwidth data transfer between two uncoupled mobile electronic devices, the method comprising the steps of:

    - displaying, on a screen of a first mobile device, a sequence of data frames, wherein each frame encodes a portion of a data set into a two-dimensional grid of channel elements, and wherein each channel element comprises a plurality of physical pixels all set to a substantially uniform emission characteristic;
    - capturing, using an image sensor of a second mobile device, said sequence of data frames over a free-space optical path; and

- decoding, by a processor on the second mobile device, the captured sequence of frames to reconstruct the data set.

2. The method of claim 1, wherein the emission characteristic is color.

3. The method of claim 1, further comprising an initial step of displaying a synchronization pattern on the screen of the first device and capturing said pattern with the image sensor of the second device to establish data transfer parameters.

4. The method of claim 3, wherein the synchronization pattern is a QR code containing at least one of: the dimensions of the channel element grid, the data rate, or an error correction scheme.

5. The method of claim 1, wherein the size of the channel elements is dynamically adjusted based on the distance or alignment between the two devices.

6. A system for high-bandwidth data transfer, comprising:

   - a first mobile device configured to:
     - encode a data set into a sequence of frames, wherein each frame is a grid of logical channel elements that are larger than the physical pixels of a display screen; and
     - display said sequence of frames on the display screen;
   - a second mobile device configured to:
     - capture the sequence of frames using its camera; and
     - decode the sequence of frames to reconstruct the data set.

7. The system of claim 6, wherein the first mobile device is further configured to initiate the transfer by displaying a machine-readable synchronization pattern.

8. The system of claim 6, wherein the data transfer rate is at least 1 gigabit per second.

9. A non-transitory computer-readable medium containing instructions that, when executed by a processor on a mobile device, cause the device to perform the steps of:

   - receiving a data set to be transmitted;
   - generating a sequence of visual data frames, wherein each frame represents a portion of the data set encoded as a grid of colored blocks, each block comprising a plurality of physical screen pixels; and

- causing a display screen of the mobile device to render the sequence of visual data frames in rapid succession.

10. The medium of claim 9, wherein the instructions further cause the device to first display a synchronization pattern containing metadata required for a receiving device to decode the subsequent data frames.

Drawings

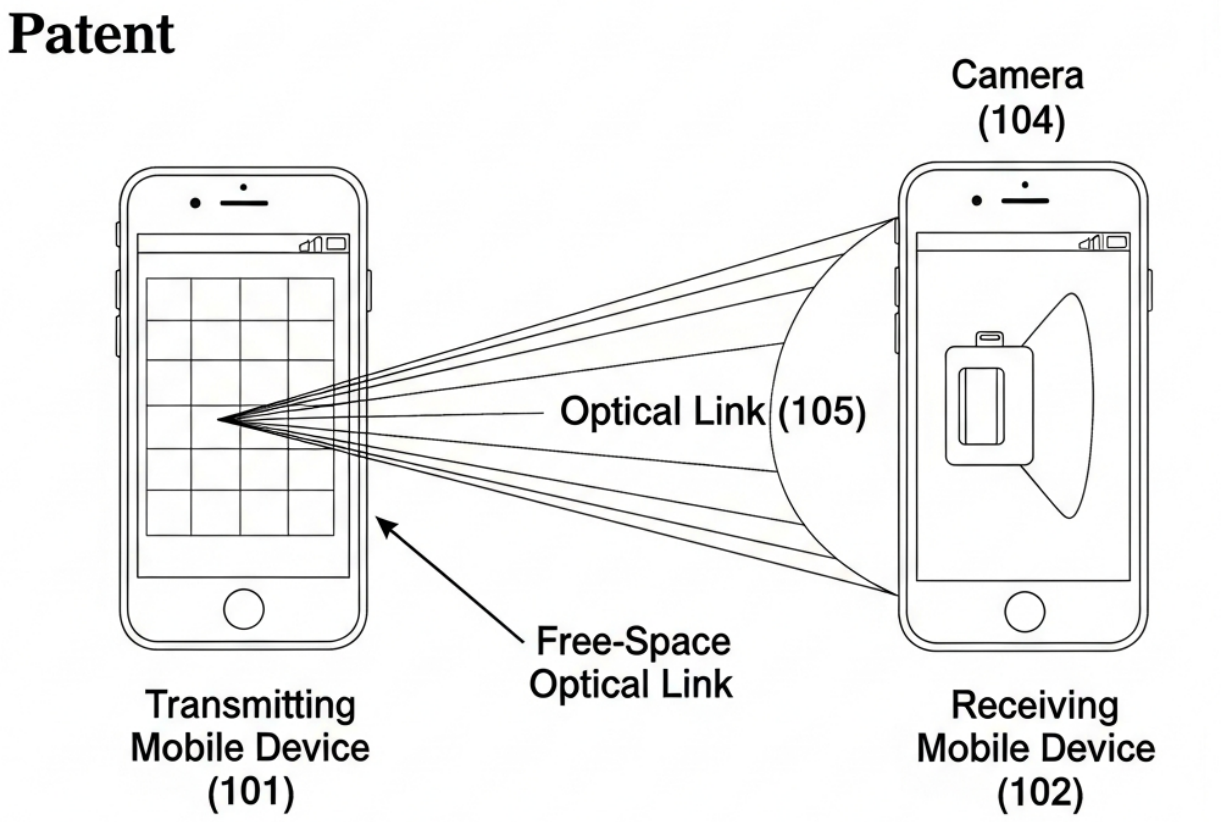


Figure 1: A conceptual diagram of the system in operation, showing two mobile devices establishing a free-space optical data link.

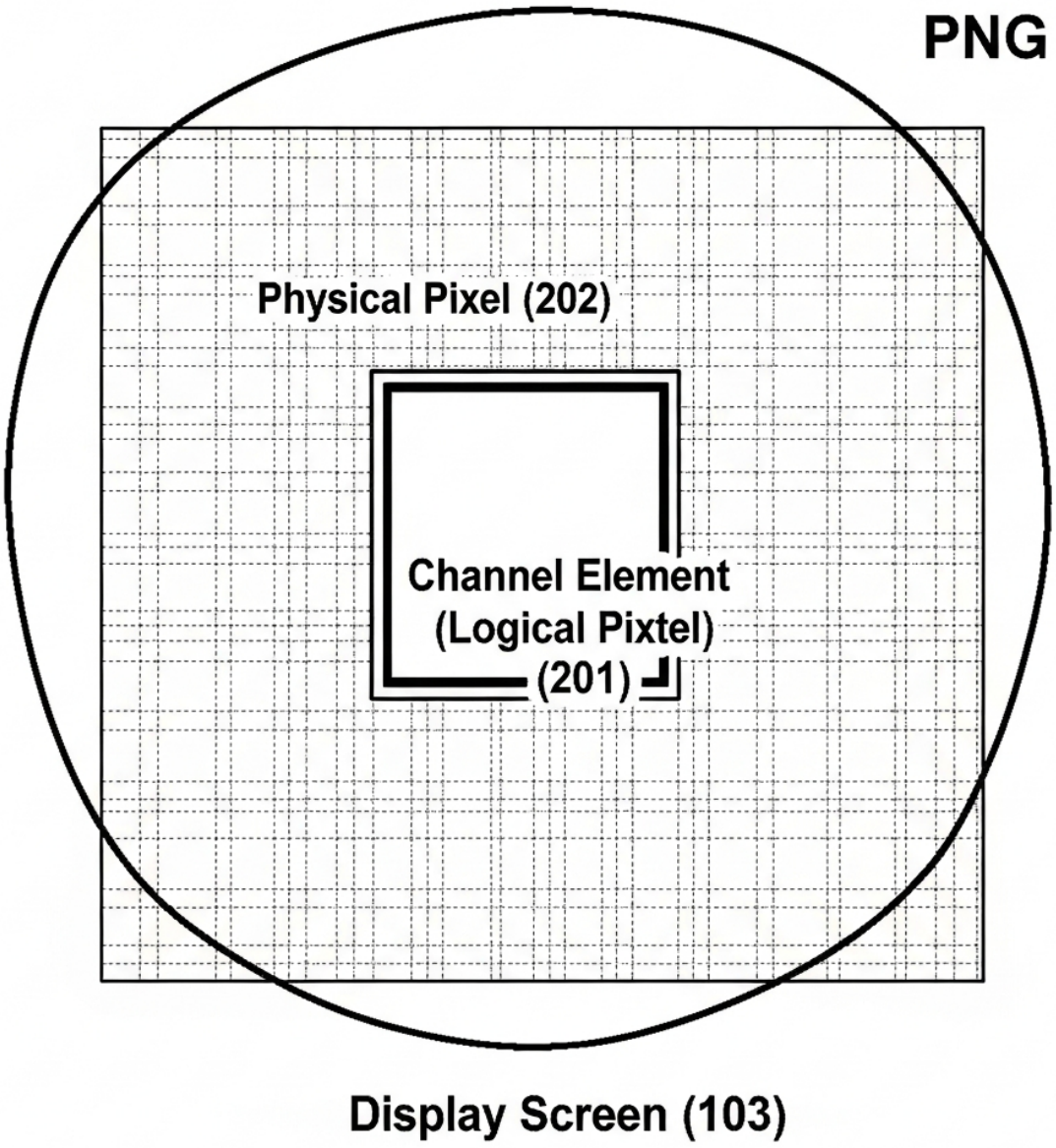


Figure 2: A diagram illustrating the concept of a "channel element," where multiple physical screen pixels are grouped into one logical data channel.

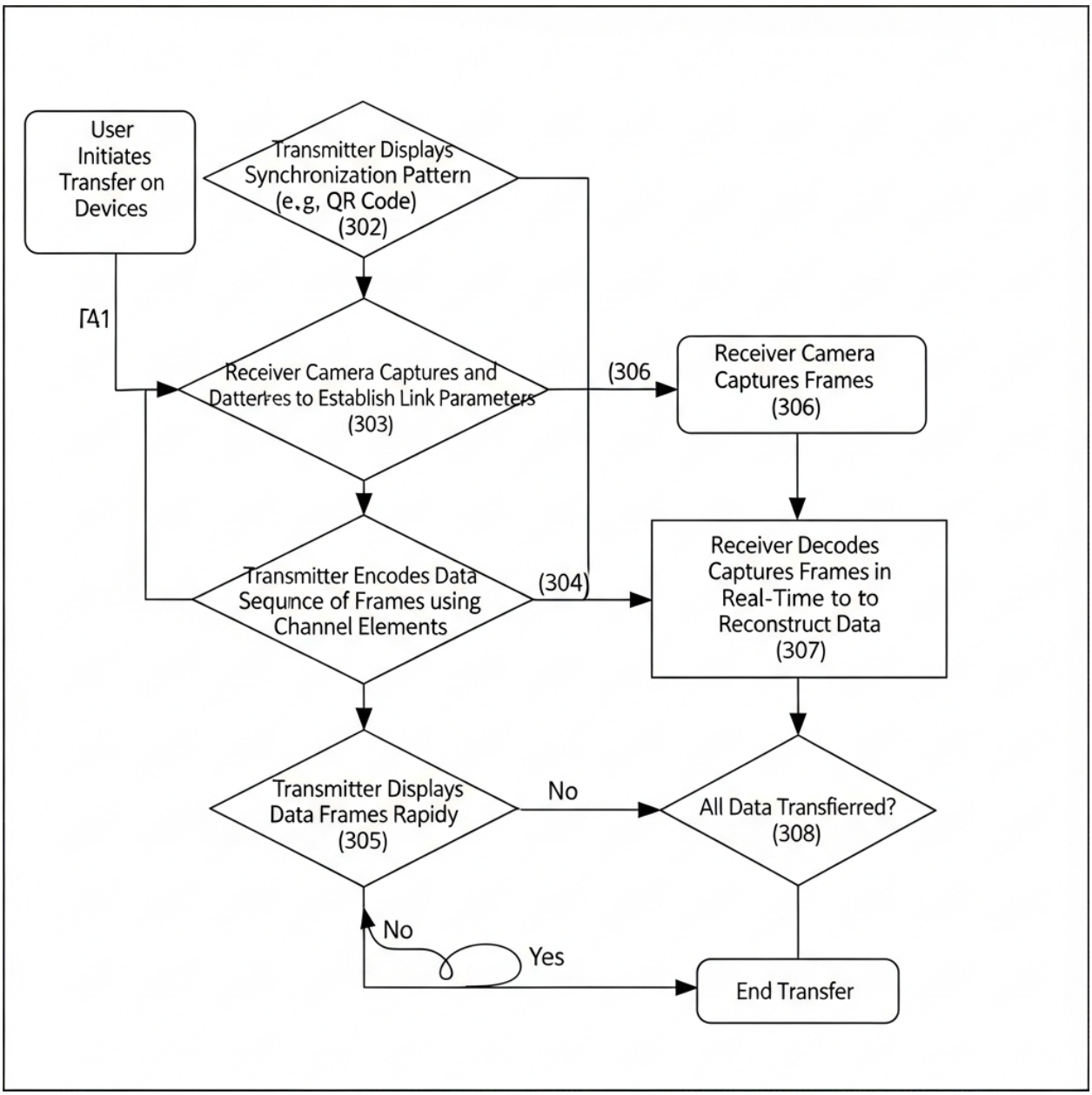


Figure 3: A flowchart of the data transfer process, from the initial handshake to the completion of the data transmission.